

My formalization project

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Chapter 1

Natural Numbers

Definition 1.

Definition 2. Natural number addition.

Theorem 3. For any natural number a , $0 + a = a$, where $+$ is theorem [2](#).

Proof. The proof follows by induction. □

Theorem 4. For any natural numbers a, b , $(a + 1) + b = (a + b) + 1$.

Proof. Proof by induction on b . □

Theorem 5. For any natural numbers a, b , $a + b = b + a$.

Proof. The base case follows from theorem [3](#).

The inductive case follows from theorem [4](#). □